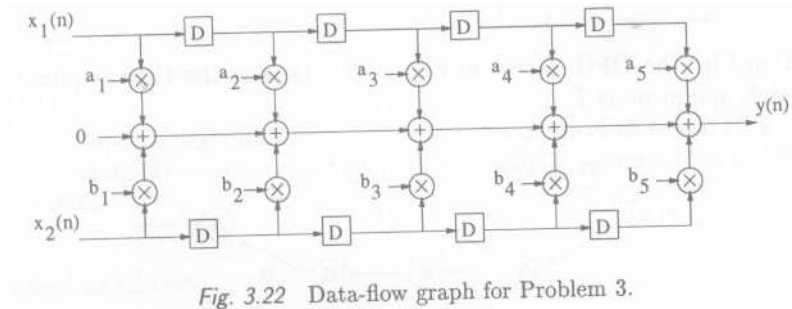


2018 NCHU EE VLSI DSP Homework Assignment # 4

Due date: 2018/5/29

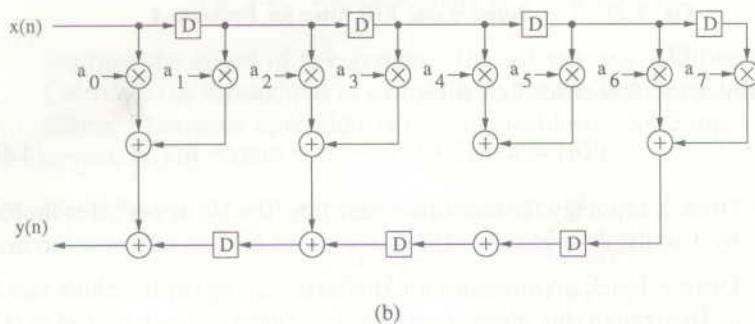
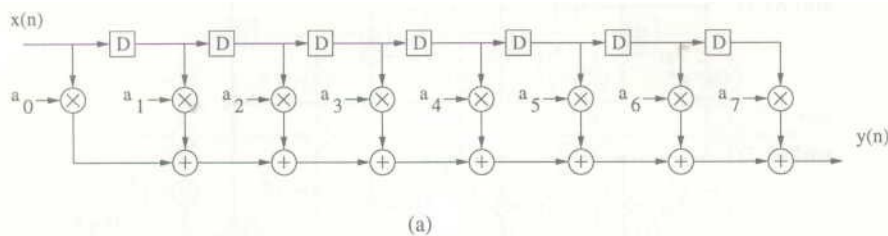
Q1. Consider the DFG shown in Fig. 3.22,

- find an equivalent data-broadcasting structure of the design, i.e. no delay line for $x_1(n)$ and $x_2(n)$.
- Compare the critical path delays of the two designs



Q2. For the two implementations of an 8-tap FIR filter below, assume the propagation delay of a multiplier is twice that of an adder, i.e. $T_m = 2T_a$. Also assume the capacitance of a multiplier is 10 times that of an adder, i.e. $C_m = 10C_a$. Design (a) is the original design version operating with $V_{dd} = 4V$, $V_{th} = 0.5V$, and clock cycle $T_s = T_m + 7T_a$ (the critical path delay). Design (b) is the low power version design operating at the same speed but with a lower V_{dd} . Note that the value of V_{dd} should not be lower than 1.2V for an acceptable noise margin.

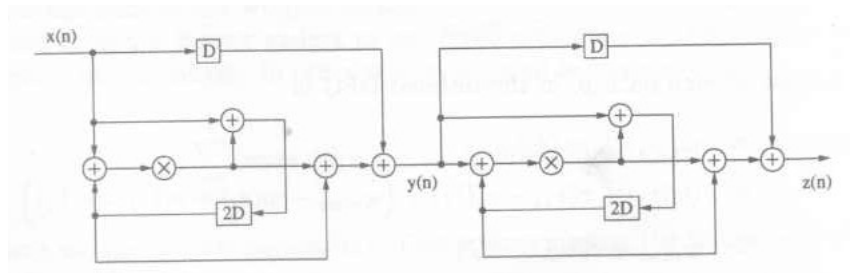
- Please calculate the V_{dd} scaling factor β
- Please calculate the % of power saving compared with the original design



Q3. Consider the DFG shown below. Assume that each multiply requires 20ns and each addition requires 8ns.

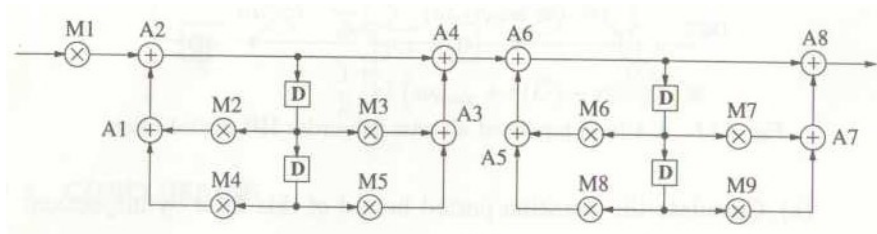
- Calculate the iteration bound of the filter

- b) What is the critical path
- c) Manually pipeline and/or retime this filter to achieve a critical path equal to the iteration bound

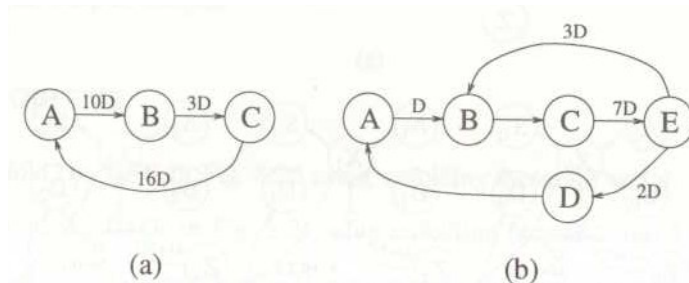


Q4. The DFG shown below describes a 4th order IIR digital filter implemented as cascade of two 2nd order sections. Assume each multiply require 2 u.t. and add requires 1 u.t.

- a) What is the critical path of the DFG? What is the iteration bound of the DFG?
- b) Retime and pipeline the DFG to minimize the clock period. What is the minimum achievable clock period obtained with pipelining and retiming?



Q5. Unfold the DFG shown below using unfolding factors 2 and 5



Q6. For the DFG shown below, assume the computing times of multiplier and adder are 3 u.t and 1 u.t., respectively. Unfold the design by a factor of 2 and apply retiming to achieve a critical path delay of 6 u.t..

